

CRIMINAL INVESTIGATION TRACKER WITH SUSPECT PREDICTION

A PROJECT REPORT

Submitted in partial fulfillment of the
Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

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MUMBAI, 400057

MAHARASHTRA

2025-2026



Parle Tilak Vidyalaya Association's

M.L.DAHANUKAR COLLEGE OF COMMERCE

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DEPARTMENT OF INFORMATION TECHNOLOGY

APPROVAL OF PROJECT PROPOSAL

PRN No. _____

Roll No. _____

Name of the Student: _____

Title of the Project: _____

Name of the Guide: _____

Is this your first submission

Yes ☐

No ☐

(Signature of Student)

(Signature of Guide)

(COORDINATOR)



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DEPARTMENT OF INFORMATION TECHNOLOGY

C E R T I F I C A T E

This is to certify that the report of project entitled

undertaken at M.L. Dahanukar College of Commerce by

_____ Seat no. _____

*in partial fulfillment of B.Sc.(I.T) Degree (Semester. V)
Examination (2025-26) has not been submitted for any other
examination and does not form part of any other course undergone
by the candidate.*

*It is further certified he /she has completed all required phases of
the project.*

Internal Examiner

External Examiner

Coordinator

College Seal

ROLES AND RESPONSIBILITIES

I Shravani and Neha equally take the responsibility of completing the project, referring that we both are the team members for this project. All the roles and responsibilities will be fulfilled by us in the given timeframe.

Role and Responsibilities for Neha

Frontend: The front-end will consist of the user interface of several modules of the applications.

This will be developed using Java

Role and Responsibilities for Shravani

Backend: The back-end will involve the overall development of the website which will be developed using MongoDB and Java respectively

Interfaces:

- Login and Authentication Page
- Case Management Page
- Admin Page
- Evidence Management Page
- Pattern Analysis Page
- Suspect Prediction Page
- Alert and Notification Page
- Investigator's Access Page
- Report Generation Page

Modules:

- Centralized Case Management System
- Evidence Management
- AI- Based Suspect Prediction
- Forensic and Biometric Integration
- Geolocation and Crime Mapping
- Investigator Access Control

DECLARATION

I hereby declare that the project entitled, **“Criminal Investigation Tracker with suspect Prediction”** done at **M.L. Dahanukar College**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** to be submitted as final semester as part of our curriculum.

Shravani Parmanand Joshi

ABSTRACT

The traditional method of handling criminal investigations often relies on manual processes, scattered information, and time-consuming analysis, which can hinder the timely resolution of cases. "Criminal Investigation Tracker with Suspect Prediction" is an advanced digital platform designed to address these challenges by offering a centralised system that allows investigators to efficiently manage case data, track investigation progress, and utilise AI-based suspect prediction. The platform leverages machine learning algorithms to analyse historical records, behavioural patterns, and related data to suggest possible suspects, thereby assisting officers in making informed decisions. It features a user-friendly interface for smooth navigation, ensures the security and confidentiality of sensitive information, and provides tools such as real-time updates, intelligent search, automated report generation, and trend analysis of criminal activities. By modernising the investigative process, this system aims to improve accuracy, reduce delays, and minimise reliance on traditional manual methods.

Keywords: Crime, Investigator, Criminal, Evidences, Suspect, Case-Reports

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CHAPTER 1

INTRODUCTION

1.1 Background

When any case is registered the investigator has to start from scratch, right from the evidence collection, crime location and eyewitness information. All it requires is the lots of time and pre-research of the case history. In today's era Traditional investigation processes are being carried out by manual tracking, with the rise of data science and machine learning there are growing opportunities to enhance through intelligent systems.

This website would initially help investigator to get all the information related to their investigation and would provide clear data without complexity. It would also predict the suspect based on the available records with the help of machine learning algorithms and data mining techniques. This would eventually help investigators to speed up the process of investigation. Also it recommends suggestions as per the crime patterns. This project majorly aims to bridge the gap between traditional and modern investigation techniques

1.2 Objectives

- To analyse entered case information and compares it with past data to suggest likely suspects with Suspect prediction module.
- To visualize where and when crimes occurred, helping spot patterns and clusters them by Crime mapping and timeline tools.
- To provide a user-friendly interface for investigators, allowing them to update, view, and analyse cases from both desktop and mobile devices.
- To ensure secure access to sensitive information through login systems and user role management.
- To Visualize investigation data (e.g., crime locations, timelines, and suspect networks) to help law enforcement see the bigger picture.

1.3 Purpose, Scope and Applicability

1.3.1 Purpose

The main purpose of this project is to support the law enforcement agencies, by providing them with smart and organized way to track ongoing criminal investigations. The project would aim to make the investigation process faster and efficient by keeping the past history records of each case in a centralised format also it uses intelligent prediction methods to suggest possible suspects based on their behaviour and crime patterns. This would help the investigator to focus their efforts, reduce time spent on manual analysis and increase the chances of solving the case more accurately and efficiently.

1.3.2 Scope Methodology

The development of the project involves a structured approach, by combining the features like data analysis, machine learning for pattern recognition. The process mainly begins with data collection which may consist of suspect profiles, crime scenes and forensic details gathered from verified sources. The main functions would be AI powered suspect prediction, crime scene mapping, evidence management system and many more. Diagrams and flowcharts were created to outline system flow and user interactions. We collected anonymized historical crime data, including crime type, location, and suspect profiles. The data was cleaned and pre-processed for use in training machine learning models. The best-performing model was selected based on accuracy and its ability to generalize to new cases.

Assumption

- This website would assume accurate and complete data.
- It assumes the system learns from past crimes to help solve new cases.
- We assume sensitive information will be kept secure and private.
- Our assumption is that the system will assist – not replace- human investigators.
- It is assumed that police officers or investigators using the system have basic computer literacy to operate the interface.

Limitations

- Depends heavily on Data Quality, as if the input is incomplete or incorrect the model's prediction and suggestion would be misleading.
- Needs regular updates and retraining as crime patterns and criminal methods evolve.
- Prediction is not always 100% accurate as the system relies on historic data and machine learning, which means predictions are based on patterns and not concrete proof.

1.3.3 Applicability

- **Faster Justice for Victims:** By predicting suspects and accelerating investigations, it can bring closure to families quicker.
- **Supporting Underserved Communities:** In areas where manual tracking is inefficient, digital tracking ensures no case is neglected or lost.
- **Real-Time Updates & Collaboration:** Investigators can share case updates, forensic reports, and evidence instantly, reducing duplication and delays.
- **Legal & Ethical Data Handling:** Tracks who accessed what data, ensuring transparency and integrity in criminal justice systems.
- **Improved Coordination:** Allows multiple officers or departments to access and update case details, improving teamwork and communication.

1.4 Achievements

Developing the “Criminal Investigation Tracker with Suspect Prediction” will enable me to gain a deeper understanding of how real-time data processing, pattern recognition, and predictive algorithms can assist in solving real-world problems. Through this project, I will enhance my practical skills in app development, database integration, and the use of machine learning techniques for suspect profiling. Additionally, I will improve my ability to design secure, user-friendly interfaces and manage sensitive data responsibly. This project will also strengthen my problem-solving and analytical skills, as I will need to interpret criminal patterns, optimise system performance, and ensure reliable outcomes. Overall, completing this project will be a significant achievement, showcasing my ability to apply technology for social good and contributing towards smarter, tech-driven law enforcement solutions.

1.5 Organisation of Report

- **Chapter 2: Survey of technologies**

This chapter provides a comparison of various programming languages and technologies. we also explain which specific technologies and programming languages we have chosen for this project and provide clear reasons for selecting them.

- **Chapter 3: Requirement and analysis**

This chapter will clearly explain the problem the project aims to solve. It will also cover the planning, scheduling, and detailed analysis of the project. This section will introduce the conceptual models, including the Use Case Diagram, an ER Diagram and an Activity Diagram, which help in understanding the system's structure and flow.

- **Chapter 4: System Design**

This chapter includes the layout of the project and information about the user interface.

CHAPTER 2

SURVEY OF TECHNOLOGIES

2.1 Available Technologies

For a complex website like “Criminal Investigation Tracker with AI-powered prediction”, it’s best to separate technologies into three distinct layers: The front end (what the user sees), the back end (The engine that powers it), and the database (where information is stored).

Front-end development (user interface) HTML builds the page structure, CSS adds styling, and JavaScript makes things interactive. To create smooth and responsive interfaces, developers often use frameworks like React, which are full-featured and structured. while Java and Python is a great backend choice, it handles data processing and run AI models. Using frameworks like .NET and Flask, you can build secure APIs, manage case data, and power machine learning predictions to support investigation decision-making.

For storing data, MySQL works well for structured data like case files using SQL. MongoDB is better for unstructured data like interview notes or images, using its own flexible query language called MySQL.

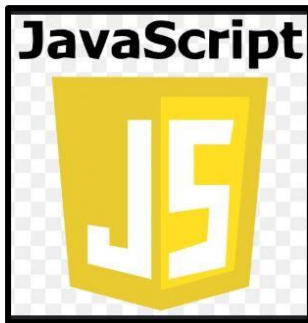
2.2 List of Technologies

The following are the technologies and software used for the development based on the particular requirements



- **HTML**

The standard mark of language for documents intended to be displayed and viewed online while a browser is also used to assist in the structure of web page is called Hyper Text Markup Language. It has numerous tags because it is a markup language Text, tables, order and unordered list which structures the web page and helps in organizing content.



- **JavaScript**

JavaScript is a lightweight interpreted scripting language used widely to enhance and add content to a web page it gets used in various ways from adding simple functions or buttons to creating complex animations.



- **Java**

It uses the same OOP principle utilized by C++ offers a broad range of systems. Java codes typically run on the Java Virtual Machine JVM and translate into generic bytecode executable on any system. It works best in offering strong performance, security and scalability for managing investigation data.



- **Python**

Another language offering an OOP approach, Python is one of the easiest to use and versatile programming languages.

In Python ecosystem you'll find a rich set of tools, libraries and frameworks that powers the back-end handling, data processing, prediction models and connecting databases with front end interface.



- **.Net**

It can be used as a robust back-end framework for tracking offering powerful tools for building secure, scalable applications management with AI models. It provides APIs to connect front-end ensuring efficient handling of investigation data.



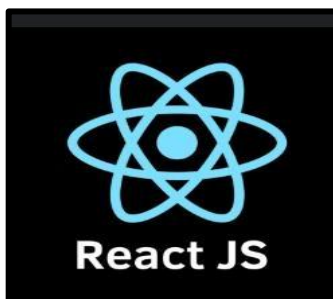
- **MySQL**

Structured Query Language SQL which is popular language for maintaining and accessing database entries. It stores structured case data like suspect profile, officer record and reports ensuring reliable, organized and efficient data retrieval.



- **MongoDB**

It is popular, open-source NoSQL database designed to store and manage large amounts of data in a flexible and scalable way. It is ideal for storing unstructured or semi-structured data. It handles evidence files, interview transcriptions, social-media data, images and flexibility with documents



- **React JS**

It is perfect for creating a dynamic and responsive user interface for the user. It allows developer to build reusable components making it easier to display and update case detail suspect profiles with smooth engaging experience throughout the investigation process.



- **Tailwind-CSS**

It offers utility-first classes that let developers quickly style components like case cards, evidence tables, and suspects profile directly in HTML.

2.3 Comparative Study

Technology	Features	Advantages	Disadvantages
Java	• Secure • Simple • Portable • Dynamic • Scalable	• Open-source • Speed • High security • Easy to use	• more memory space • Cost increases
C#	• Interoperability • Simple • Secure • Type safe	• Rich-Libraries • Easy to Learn • Scalable • Reliable	• Heavy Runtime • Requires Compilation • Runs mostly on .NET framework
Python	• Interpreted language • High-level language • Free and open source	• Easy to use • Easy to learn • Human Readable code • Secure	• High-memory consumption • Limited support • Runtime Errors
MongoDB	• High performance • Scalable • Open source • Easy-integration	• Build-in Security • Handles complex datatypes • Indexing for speed	• Data redundancy • High memory usage • Slow with large joins
MySQL	• Open source • Cross-Platform support • High-Speed and Performance	• Free and open source • Reliable • User-friendly	• No-graph database support • Limited JSON support

2.1 Comparative Study of Technologies

2.4 Selected Technology

List of selected technologies

1. Java
2. Python
3. SQL
4. Tailwind CSS

Selected technologies and why?

- **Java**

One of the key reasons to use Java is its ability to run on any device equipped with a Java Virtual Machine (JVM), which makes applications highly portable. Java offers a rich set of libraries and frameworks that simplify backend development and database integration. Additionally, Java is known for its reliability, scalability, and performance. It is on the back-end or middleware for things like managing user logins, filtering records. .

- **Python**

Python plays a crucial role in managing the backend operations, such as handling user authentication, storing case details, and processing investigation data. Using frameworks like Django or Flask, Python helps in building a secure and structured web application. Additionally, it allows integration of machine learning models through libraries to analyse crime patterns and predict potential suspects.

- **SQL**

We use SQL in our website to store, manage and retrieve structured data such as user information, case reports, and evidence details. it allows for efficient querying and data manipulation, making it easy to access specific records when needed.

- **Tailwind-CSS**

Tailwind CSS is used to design a clean, responsive, and user-friendly interface. It helps in styling the login and registration pages. The dashboard, which displays crime statistics, case summaries. It is also used to organise the case tracking pages, showing details, updates, and timelines in a structured format. Overall, Tailwind improves the frontend experience.

CHAPTER 3

REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

The current investigation system is a bit complex and consumes a lot of time for the investigator. They need to get a deep-down study of the cases by manually tracking all the registered cases. Retrieving suspects' complete background history is not always accurate and efficient, so designing a system that gives accurate and real-time data.

- **Data Quality and Availability**

The system doesn't always have all the necessary data due to incomplete crime reports, missing evidence, or inaccurate witness statements. Also, personal data about a suspect or witness should be securely managed to prevent misuse. Data often comes from multiple sources, and integrating this information into a system can be technically challenging.

- **Ethical and Legal issues**

Using predictive models in criminal investigations brings up serious concerns about fairness and due process. One major issue is that people could be judged or treated unfairly based on an algorithm's prediction rather than solid evidence. This means that an individual could be flagged as a suspect without any real proof, simply because of the data the algorithm is trained on.

- **Privacy and Legal Issues**

Predicting suspects using AI or data analysis comes with sensitive ethical concerns. There's a fine line between helping investigations and violating someone's privacy or presuming guilt before evidence is solid.

- **Delayed access to real time data**

In many cases investigators don't get timely updates from the field if someone is arrested or new evidence surfaces it might take hours or even days to reach the main investigation team which slows down the progress.

- **Lack of user training**

Some officers might not be comfortable using new digital tools or may misuse the prediction feature without fully understanding how it works leading to flawed decision.

3.2 Requirements Specification

- **Data Quality and availability**

By creating user friendly structured forms for entering crime report, witnesses delayed an evidence log. This would ensure that no crime critical information is skipped.

- **Ethical and Legal Issues**

Addressing ethical concerns in predictive policing requires transparency and human judgment. Algorithms should be guided in such a way that it should be open to understand public and legal challenge. Technology must support justice and never replace it or undermine human rights.

- **Privacy and Legal Issues**

AI should only support investigator, not replace them. People deserve to be treated as innocent until proven guilty. Clear rules concern and transparency must guide how data is used always respecting individual rights.

- **Delayed access to real time data**

Set up a real-time update system that pushes new evidence or witness statements to relevant users instantly. This keeps investigators informed and avoids working with outdated data.

- **Lack if user training**

Officers should be trained to use digital tools wisely. By understanding how prediction works they can make fair informed decisions using technology as support and not a shortcut for justice.

3.3 Planning and Scheduling

Gantt Chart:

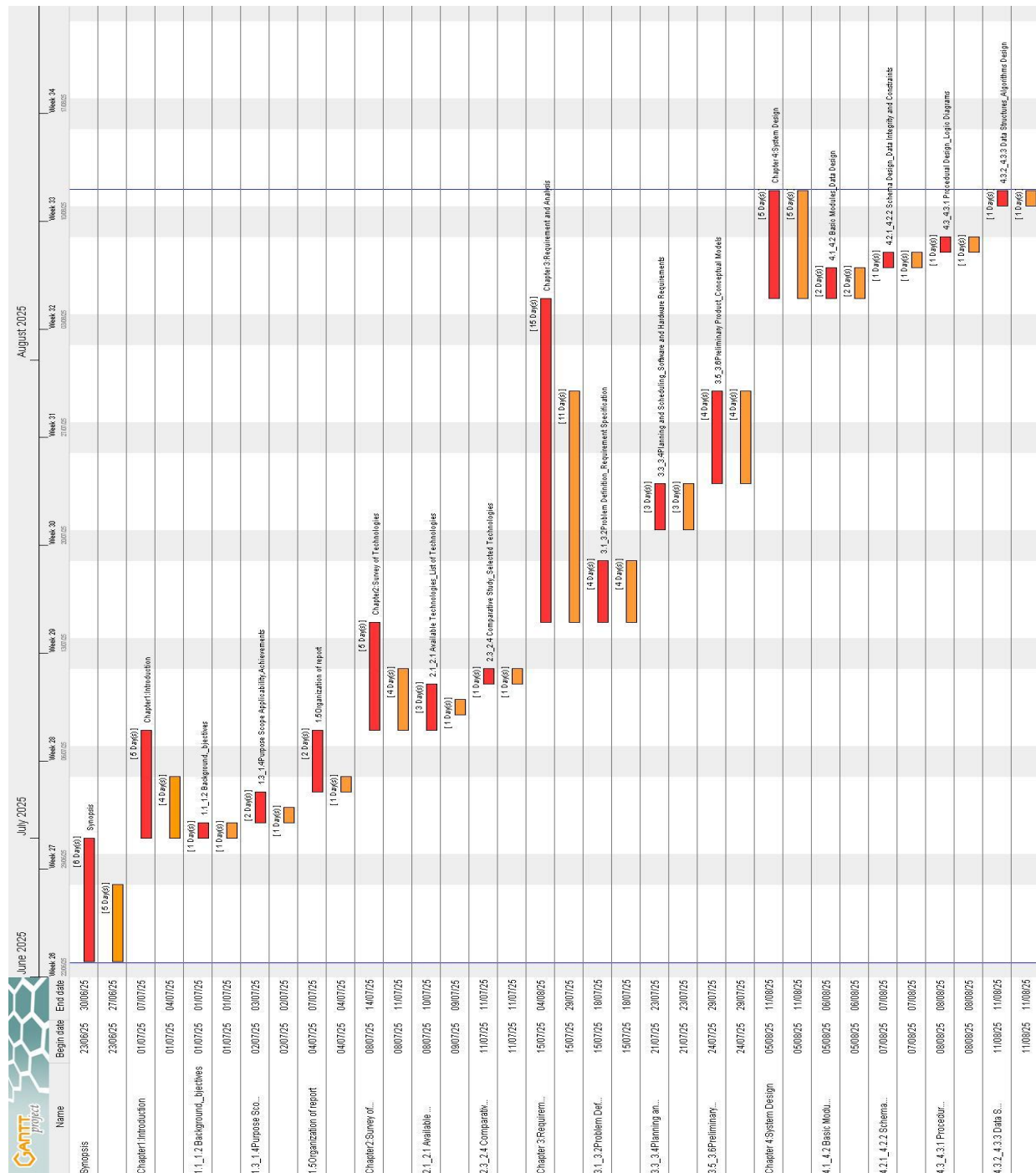


Figure 3.1 : Gantt Chart

A Gantt chart is a type of bar chart that visually represents a project schedule overtime. It is commonly used in project management to plan, schedule and track task and activities within a project

Pert Chart:

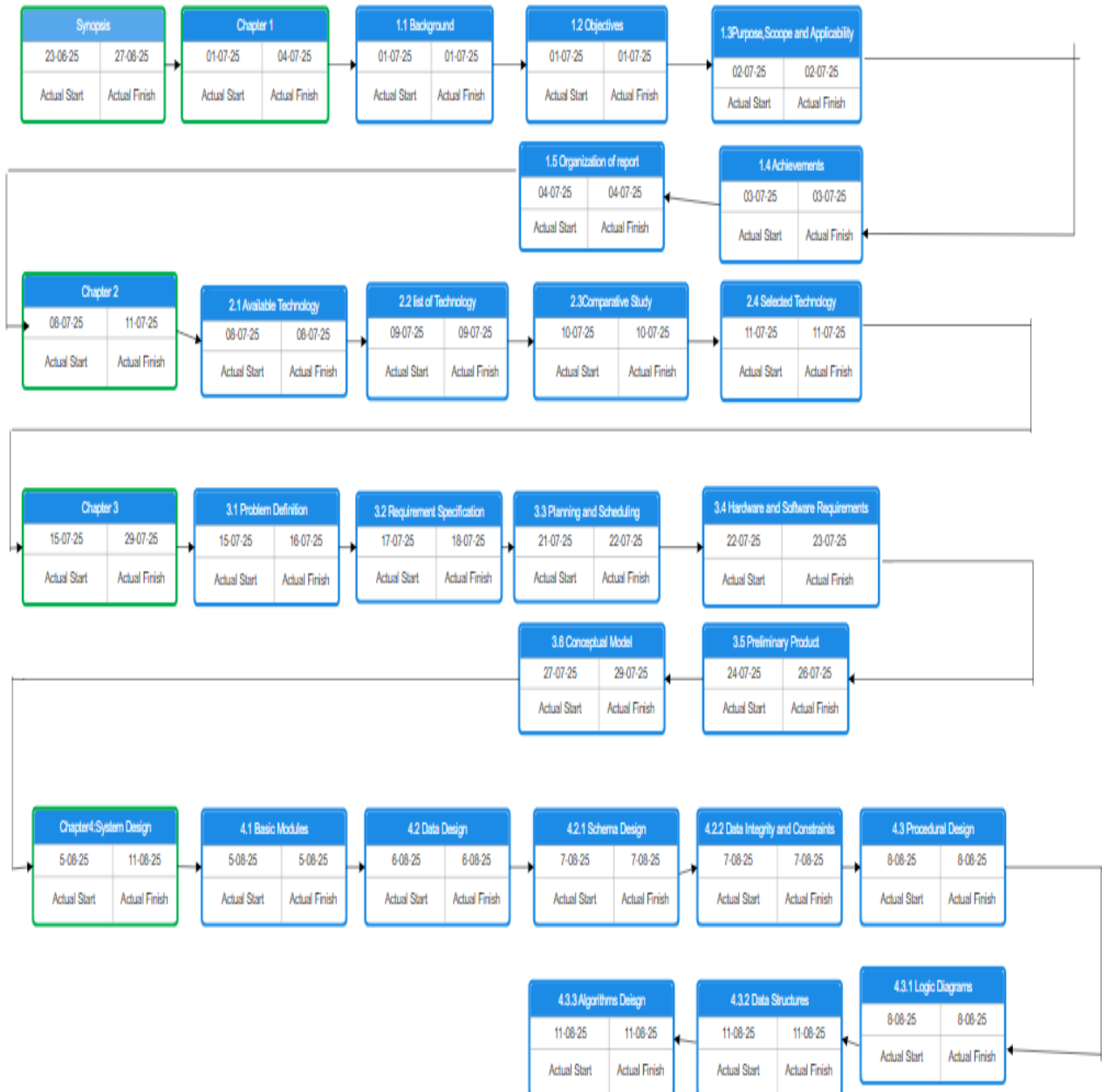


Figure 3.2 : PERT Chart

A Pert Chart program evaluation and review technique. Chart is a project management tool used to plan. Schedule and central complex project. It provides a graphical representation. of project, task and their independence. Helping to analyse and visualize the sequence of activities required to complete the project.

3.4 Software and Hardware Requirements

The Hardware and Software requirements for the Project “Criminal Investigation Tracker” website can vary based on factors. Such as scale of the website, Feature it offers and performance of the system. Below are some general hardware and software requirements as per user as well as for developers.

3.4.1 Developer requirements:

1. Hardware requirements

- Ram should be a minimum of 8GB, and 16GB or more is recommended.
- Storage is 256GB SSD is the minimum, while 512 GB SSD is preferable for faster operation
- A stable internet connection is also essential for accessing online resources

2. Software requirements

- Backend technologies: Python and Node.js
- Front-end technologies: HTML, Java
- Database: MySQL and MongoDB

3.4.2 User requirements

1. Hardware requirements

- Device: Laptop, desktop or smartphone
- Minimum RAM: 4GB
- Storage 100-200MB

2. Software requirements

- Operating system: Windows, macOS, Linux
- Web browser: the latest version of Chrome

3.5 Preliminary Product Description

The Project “Criminal Investigation Tracker with Suspect Prediction” is a modern tool designed to support law enforcement and investigative terms. It is used for managing complex cases more efficiently. Investigation often involves vast Disconnected data from witness-

account and case-files to forensic reports and surveillance logs. This platform brings everything together in a centralized dashboard, making it easier, organize, visualize and track progress in real time. More than just a digital filing system, It uses machine learning to analyze historical and ongoing data, identifying patterns that suggest likely suspect.

3.6 Conceptual Model

3.6.1 Use Case Diagram

A Use Case Diagram is the Unified Modeling language (UML) is a type of behavioral diagram defined by and created from a use case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actor, their goals and dependencies between those cases. The main purpose of use case diagram is to show what system functions are performed for which. Actor depicts their role.

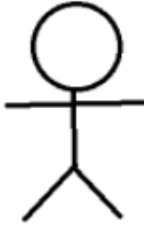
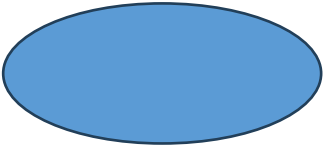
Component name	Symbols	Description
Actor		An actor specifies a role played by a user or any other system that interacts with the subject.
Use Cases		Represents the different uses that a user might have. It denotes the system's behavior or a service provided by a system.
Association		A line between actor and use cases That indicates a third participant or interacts with use cases.

Table 3.1: Symbols in use case diagram.

Actors with respect to project:

- Investigator
- Police officer
- Authorities/Prosecutors
- System Administrator



Figure 3.3 Use Case Diagram

3.6.2 Activity diagram.

Activity diagrams are graphical representation. Of workflow of stepwise activities and actions with support for choice, iterations and concurrency. The Unified Modeling Language. Activity diagram intended to model both computational and organizational processes.

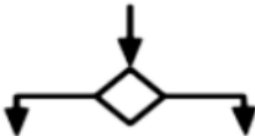
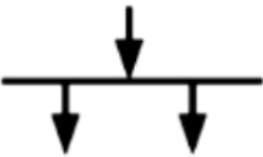
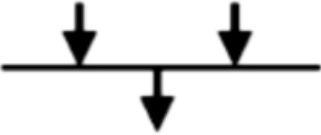
Component name.	Symbol.	Description.
Action.		These are the actions or tasks performed within the system.
Initial. State.		Activity diagram starts from this step, also known as the entry state.
Final state.		This is the last stage of the activity diagram. It is placed where activity ends.
Decision Node		It represents flow of control is determined, allowing the diagram to branch into different parts based on specific conditions.
Fork.		Two process execute. Either concurrently or in parallel. At this location. It generally includes a single input, but may or may not get one output.
Join.		A join is one where two results of concurrent activity. And form a single result.
Control flow.		Connectors between two states or two actions to depict the flow. Show the sequence of execution.

Table 3.2: Symbols in Activity Diagram

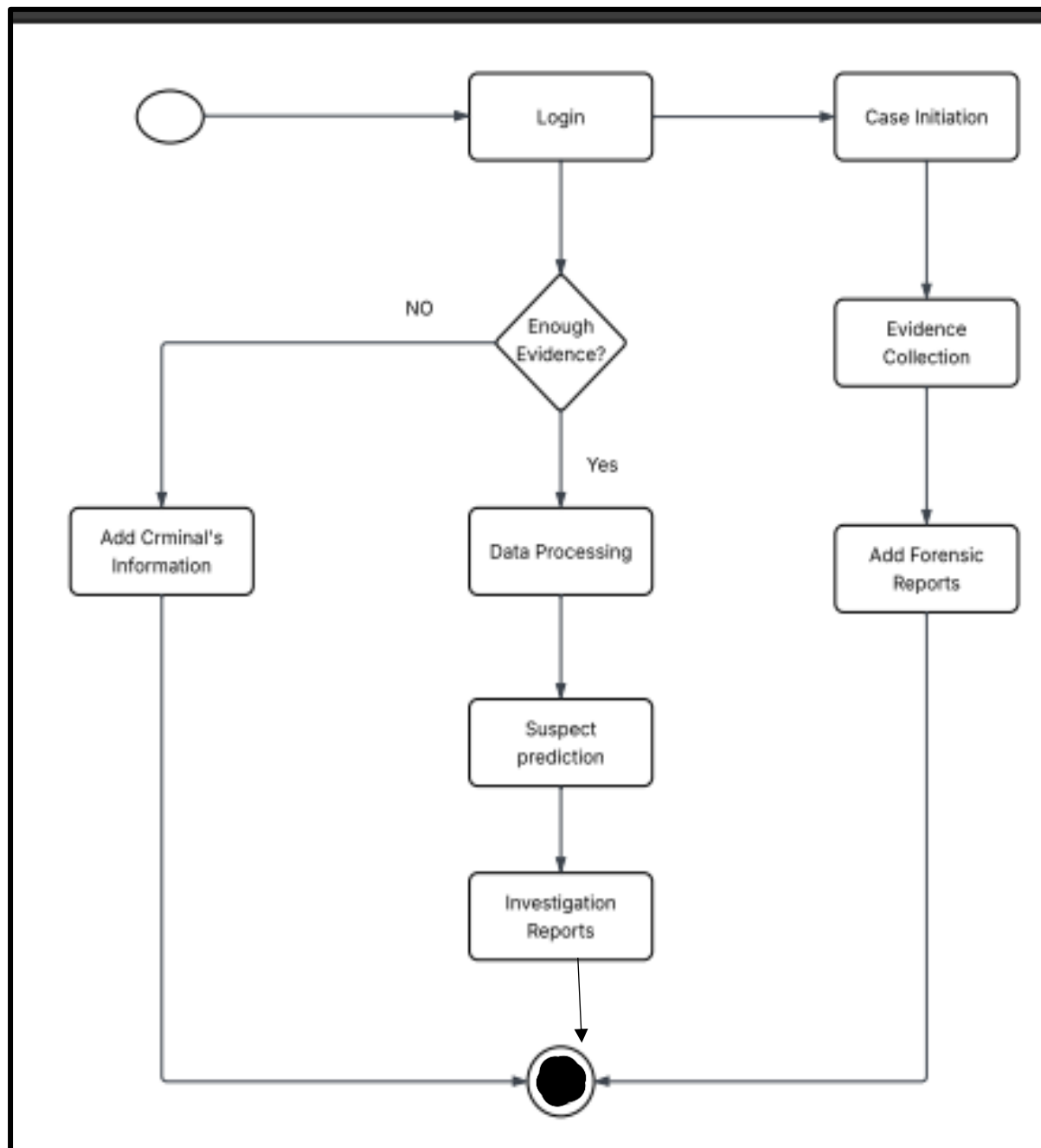


Figure 3.4 Activity Diagram

3.6.3 Data flow diagram.

A data flow diagram DFD is a visual representation in software engineering that depicts the Flow of data within a system. It provides a structured way to illustrate how data moves through various process, data stores and external entities in our system. DFDS use standardized symbol to represent process, data store, dataflow and external entities and they show how components interact to achieve specific functionalities. DFDS are commonly used to analyze, model and communicate the flow of information within a system, making them valuable tool for understanding system process and requirements during software development and design phases.

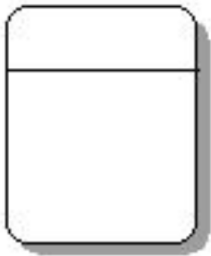
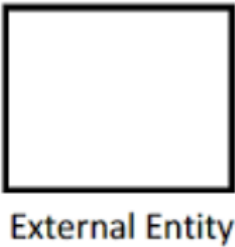
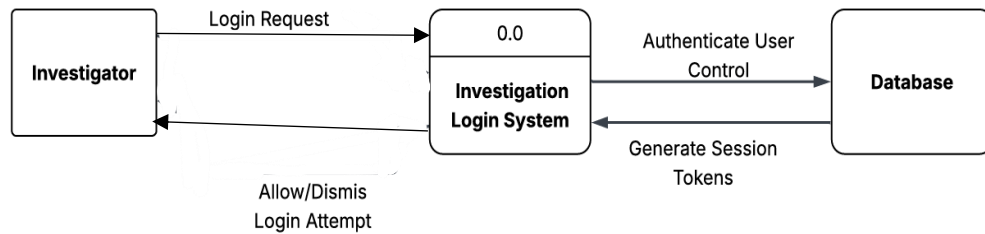
Component name.	Symbols.	Description.
Process		Process represent activities or functions that transform input data into output data. They are the actions performed on data within the system.
Data source.		Data store represents Repositories. Where data is stored. They can be database files or any other storage location.
External entity.		External entities are source or destination of data that interact with the system but are outside of it. They represent user, other system or external data source.
Dataflow.		Dataflow depicts the movement of data between process, data store and external entities. They indicate how data is transferred process. or stored within the system.

Table 3.3 Symbols used in Data flow diagram.

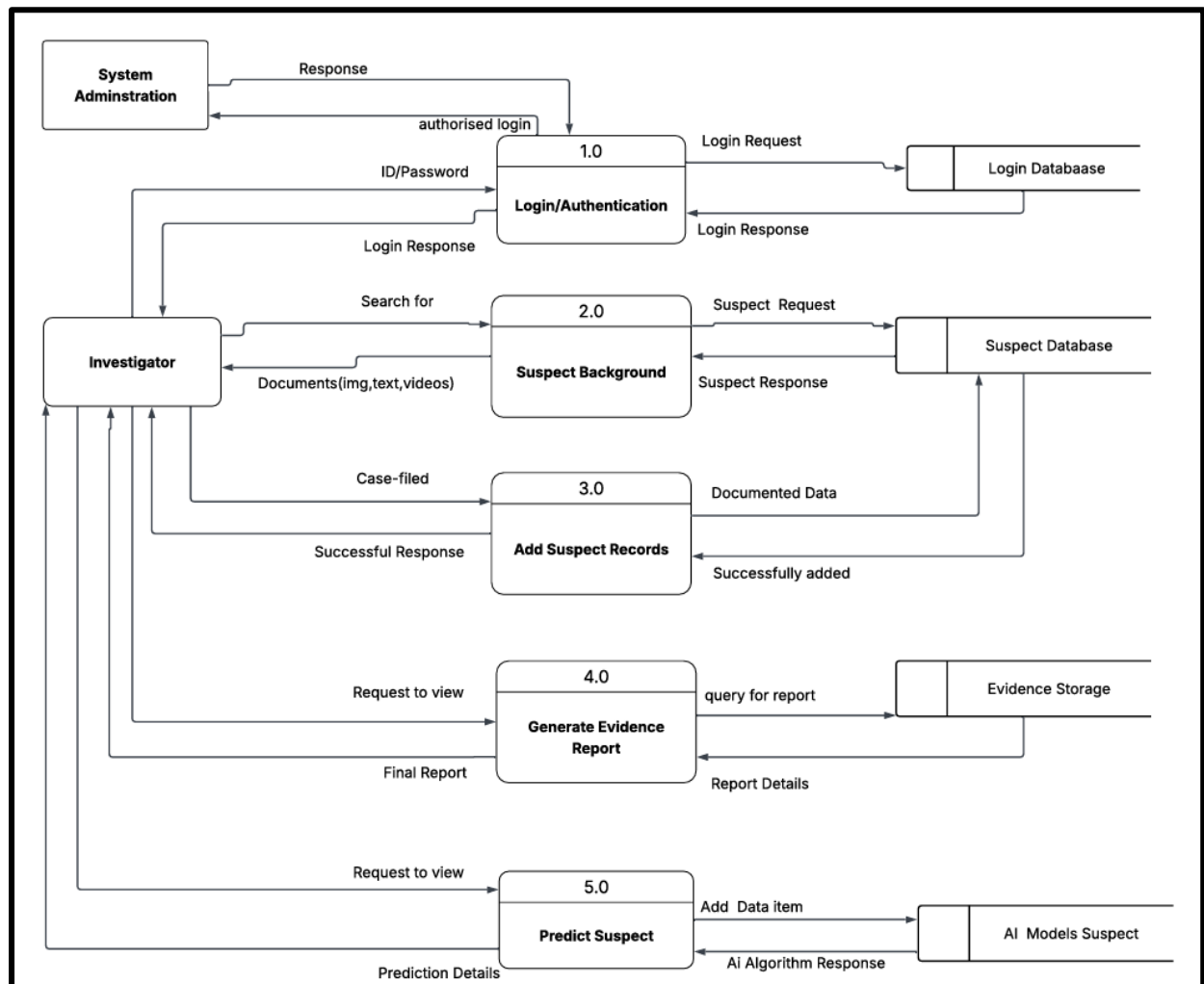
Context level DFD : level 0.

The Context level DFD data flow diagram is the most basic and top-level representation of a system. It depicts the system as a single process. Connected to the external entities by data flow. The diagram is used to showcase the interaction between system and external environment, omitting internal complexities.

Level 0 : System Login Process



LEVEL 1 DFD



LEVEL 2 DFD

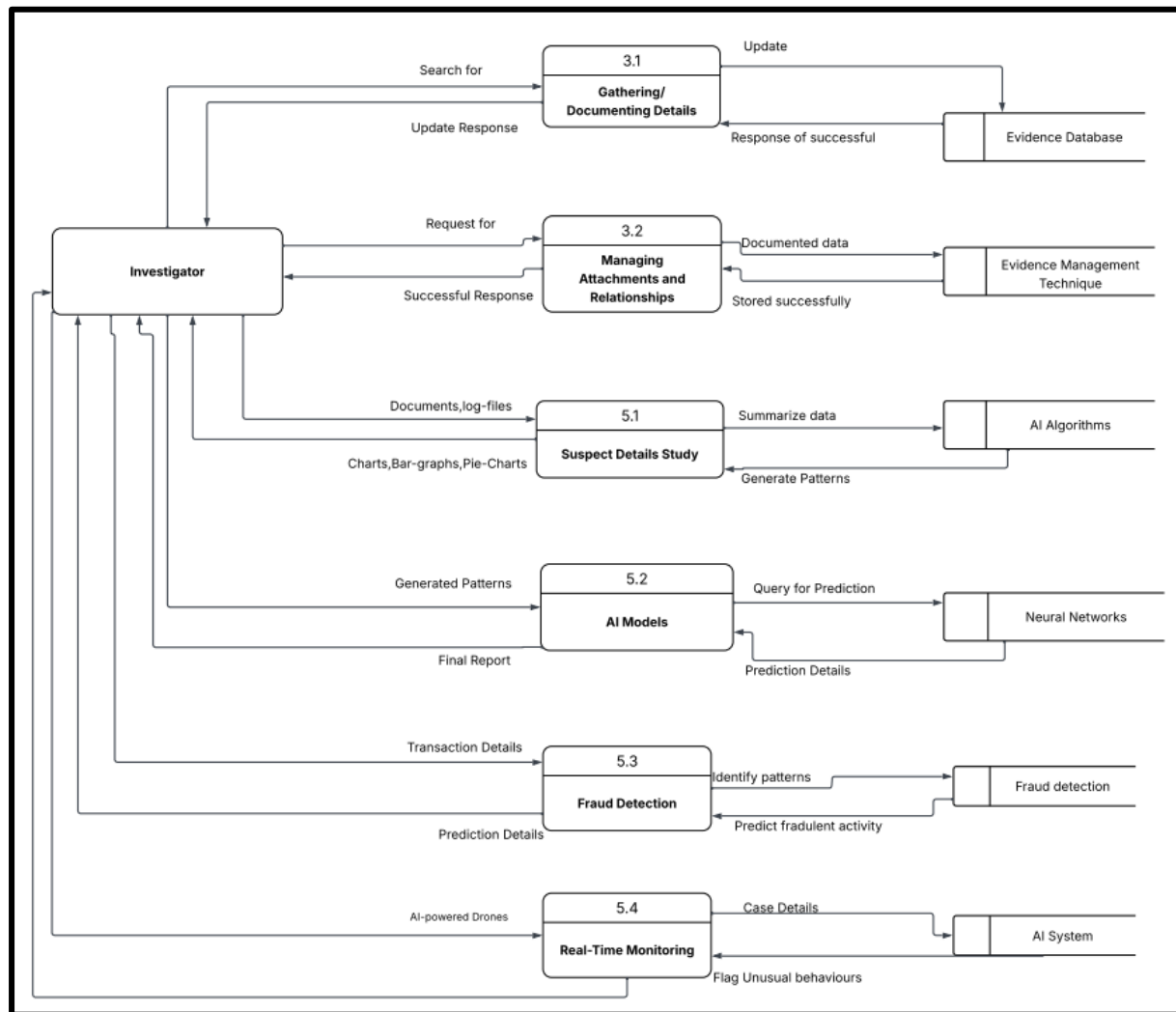


Figure 3.5 Data flow Diagram

3.6.4 ER Diagram

An entity relationship, uh diagram is a visual representation used to model the structure of a database. ER diagrams use symbols to represent entities, attributes, and relationships. ER diagrams are particularly useful. For designing and understanding database and their relationship.

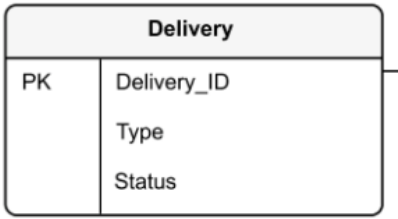
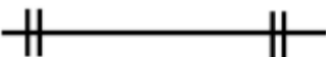
Component name.	Symbol.	Description.
Entity.		These are the tables that have data stored above them in the database.
One-to-one Relationship.		When only one instance of an entity is associated with the relationship then it is known as one-to-one relationship.
One-to-many Relationship.		When only one instance of entity on the left and more than one instance of the entity on right associates with the relationship then this is known as one to many relationship.
Many-to-many Relationship		When more than one instance of the entity on both the sides. Are associated Then it is known as many-to-many relationship

Table 3.4: Symbols in ER Diagram

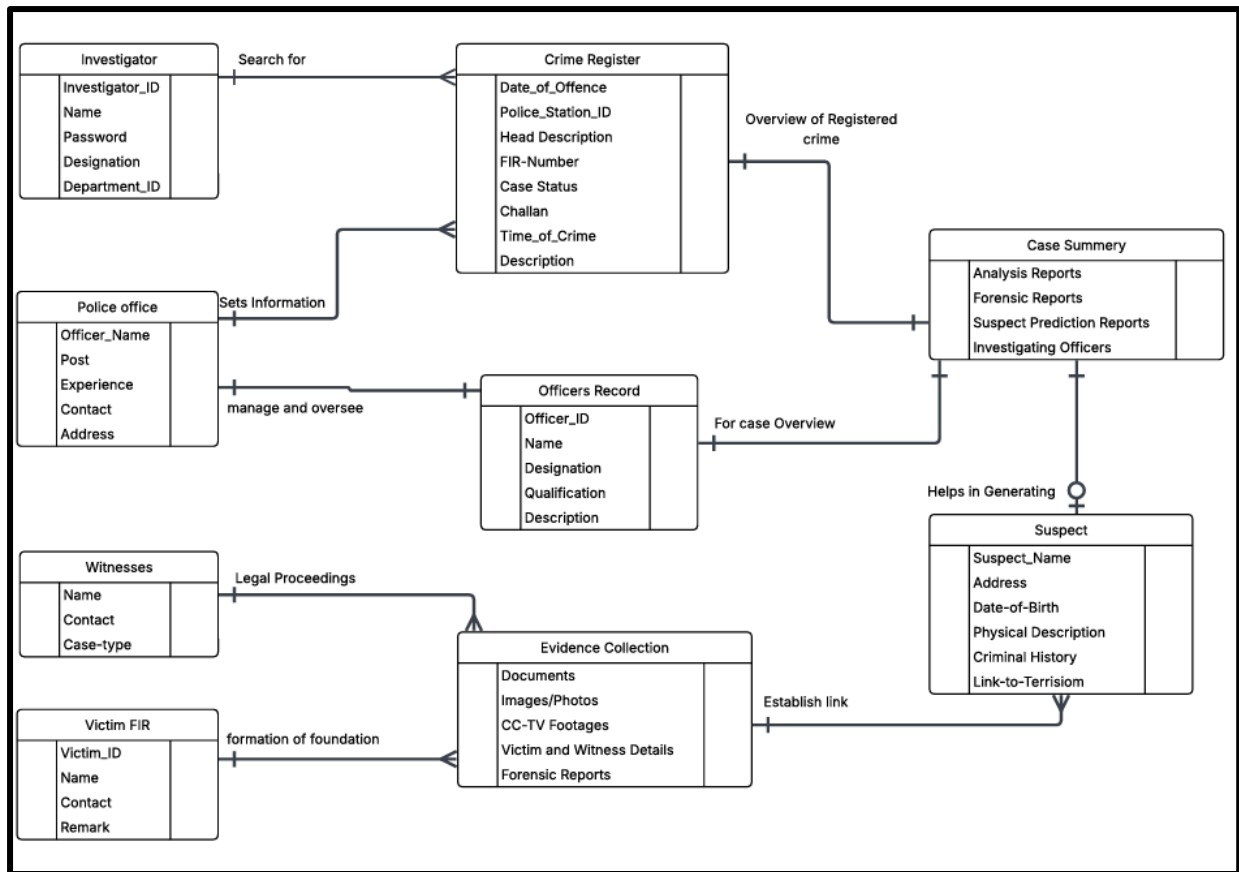


Figure 3.6 ER Diagram

CHAPTER 4

SYSTEM DESIGN

4.1 Basic Modules

In the design of a Criminal Investigation Tracker with Suspect Prediction, basic modules represent the essential components that form the complete system. By dividing the tracker into modules, investigators and developers can focus on individual parts such as user management, evidence handling, suspect prediction, and reporting independently. This modular approach simplifies system updates, supports scalability, and ensures that each function can be improved without affecting the entire application.

The use of modules also provides a clear structure, making the system easier to understand, maintain, and extend as new investigative features are required.

- **User Management Module:** Secure login, investigator registration, role-based access (admin, officer, analyst), and profile management.
- **Case Management Module:** Create new cases, view and update case details, assign cases to investigators, track case history.
- **Evidence Management Module:** Upload evidence (documents, images, audio, video), store metadata, categorize evidence, attach to specific cases.
- **Suspect Prediction Module:** Apply AI/ML models to analyze evidence, generate suspect lists with probability scores, update predictions with new evidence.
- **Investigation Tracking Module:** Assign investigation tasks, update case progress, maintain logs of activities, schedule follow-ups.
- **Reporting Module:** Generate detailed reports (case summary, suspects, evidence, progress), export reports in PDF/Word, and close case files.

4.2 Data Design

4.2.1 Schema Design

The database plays a crucial role in securely storing and managing large volumes of investigative data. A database schema acts as the blueprint of this system, defining how case details, evidence, suspects, and user information are structured and connected. Schema design is the process of planning this structure by specifying the data types, relationships between entities such as cases, suspects, and evidence, and the rules for how the data can be stored, updated, and retrieved. A well-designed schema ensures that the tracker operates efficiently, supports quick suspect prediction through AI/ML integration, and provides investigators with fast and reliable access to information.

- **Database Name:**

CrimeTrackDB

- **List of tables:**

Users

Cases

Persons

Suspect

Evidence

Interviews

4.2.2 Data Integrity and Constraints

- **Database Name:**

CrimeTrackDB

- **List of Tables**

Users

Field	Type	Null	Key	Default
id	int(11)	NO	PRI	NULL
full-name	Varchar(50)	YES		NULL
email	Varchar(100)	NO		NULL
password	Varchar(100)	NO		NULL
role	Varchar(100)	NO		NULL

Table 4.1: Users Database Table

Cases

Field	Type	Null	Key	Default
id	int(11)	NO	PRI	NULL
title	Varchar(50)	YES		NULL
incident datetime	Int(20)	YES		NULL
incident location	Varchar(100)	YES		NULL
assigned to	Varchar(100)	YES		NULL

Table 4.2: Cases Database Table

Persons

Field	Type	Null	Key	Default
Id	int(11)	NO	PRI	NULL
Name	Varchar(100)	YES		NULL
Gender	Varchar(100)	YES		NULL
date-of-birth	Int(50)	YES		NULL

Table 4.3: Persons Database Table

Suspect

Field	Type	Null	Key	Default
Id	int(11)	NO	PRI	NULL
Persons	Varchar(50)	YES		NULL
Role	Varchar(100)	YES		NULL
Case	Varchar(100)	NO		NULL
Status	Varchar(100)	NO		NULL

Table 4.4: Suspect Database Table

Evidence

Field	Type	Null	Key	Default
Id	int(11)	NO	PRI	NULL
Title	Varchar(50)	YES		NULL
uploaded-by	Varchar(100)	YES		NULL
uploaded-at	Varchar(100)	YES		NULL
case-id	Varchar(100)	YES		NULL

Table 4.5: Evidence Database Table

Interview

Field	Type	Null	Key	Default
Id	int(11)	NO	PRI	NULL
Case id	Varchar(50)	YES		NULL
Person id	Varchar(100)	YES		NULL
Conducted-by	Varchar(100)	YES		NULL

Table 4.6: Interview Database Table

4.3 Procedural Design

Procedural design organizes a system into a series of sequential steps. In a “Criminal Investigation Tracker with Suspect Prediction”, tasks like logging in, entering crime details, collecting evidence, and running AI/ML suspect prediction follow a clear, ordered workflow. Decision points, such as adding more evidence, ensure the process remains controlled, efficient, and systematic from case start to closure.

4.3.1 Logic Diagrams

It can be considered a diagram to define the relation between the software components. The main purpose of designing logical architecture is to plan for the system and used it before communication before designing and implementing the system and by this diagram, the user can get an overview of the system.

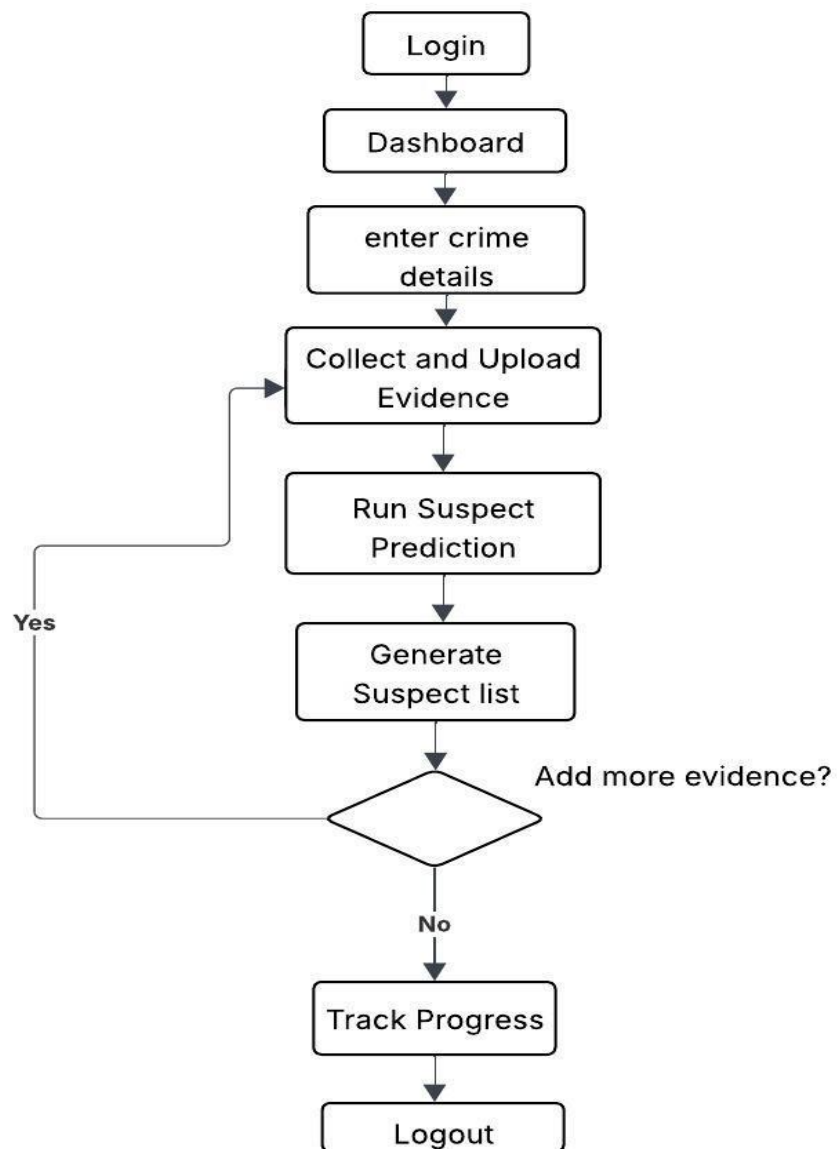


Figure 4.1: Logic Diagram

4.3.2 Data Structures

Data structures organize and manage information such as case details, evidence, suspects, and timelines. These structures allow efficient tracking of investigation progress and relationships between entities. This enables faster analysis, suspect prediction, and informed decision-making by investigators.

- **Array** – Useful for storing fixed-size lists like a list of case IDs, suspect IDs, or probability scores.
- **Linked List** – Can be used for maintaining sequential case updates or logs, where entries are added over time.
- **Stack** – Suitable for handling undo/redo operations during data entry or investigation notes.
- **Queue** – Useful for managing tasks in order, e.g., assigning investigation requests or evidence analysis jobs.
- **Tree** – Helpful for organizing hierarchical crime categories, evidence types, or decision processes.
- **Binary Tree** – Can be used in AI/ML models for decision-making.
- **Graph** – Very important here: can represent relationships between suspects, witnesses, locations, and events (crime networks).
- **Heap** – Useful for prioritizing suspects or cases (e.g., probability-based suspect ranking in priority queues).
-

I would consider data structures like:

- **Graph:** To map relationships between suspects, crimes, witnesses, and locations.
- **Heap (Priority Queue):** To rank suspects based on prediction probability.
- **Tree / Binary Tree:** For structured decision-making in AI/ML algorithms.
- **Queue:** For task assignment and investigation workflows.

4.4 User Interface Design

- **Login and Authentication**

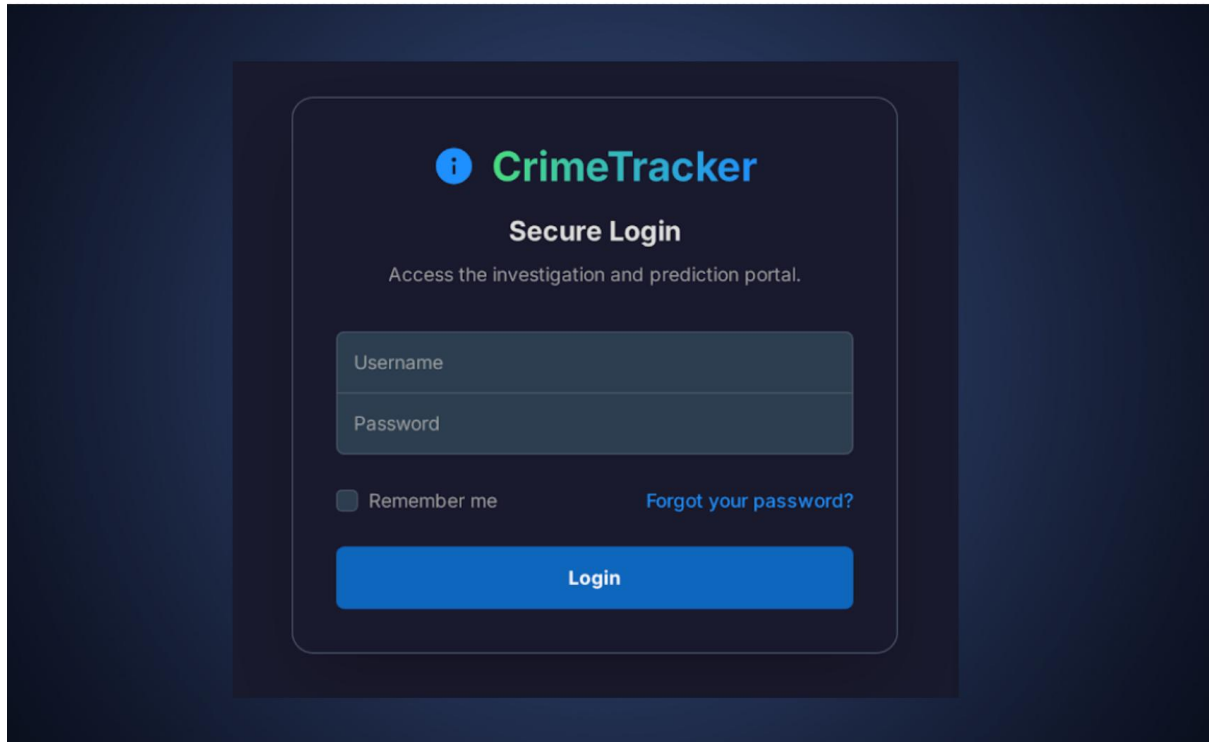


Figure 4.2: Login and Authentication Page

- Login and authentication involve users proving their identity to access secured resources (e.g., websites or applications).
- Authentication is the broader process of identity verification.
- User credentials such as username, password, biometrics, or secure ID/token are used for verification.
- Successful authentication grants the user access to the system or resource.

- **Case Managements**

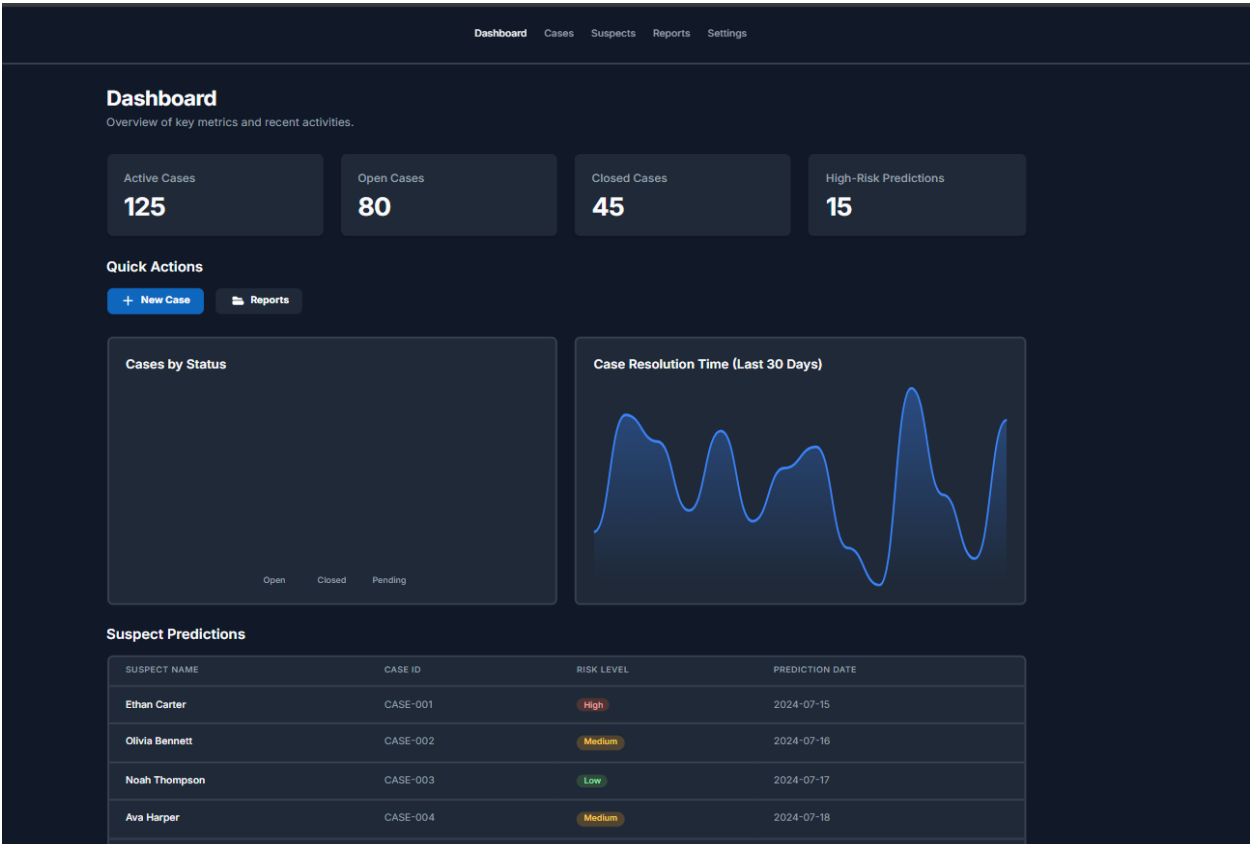


Figure 4.3: Case Management Page

- Case management is a systemic and collaborative approach.
- It plays a critical role in the public sector by handling investigations efficiently.
- Ensures thorough investigation and timely resolution of various cases.
- Promotes accountability, organization, and structured handling of case information.

- **Admin Panel**

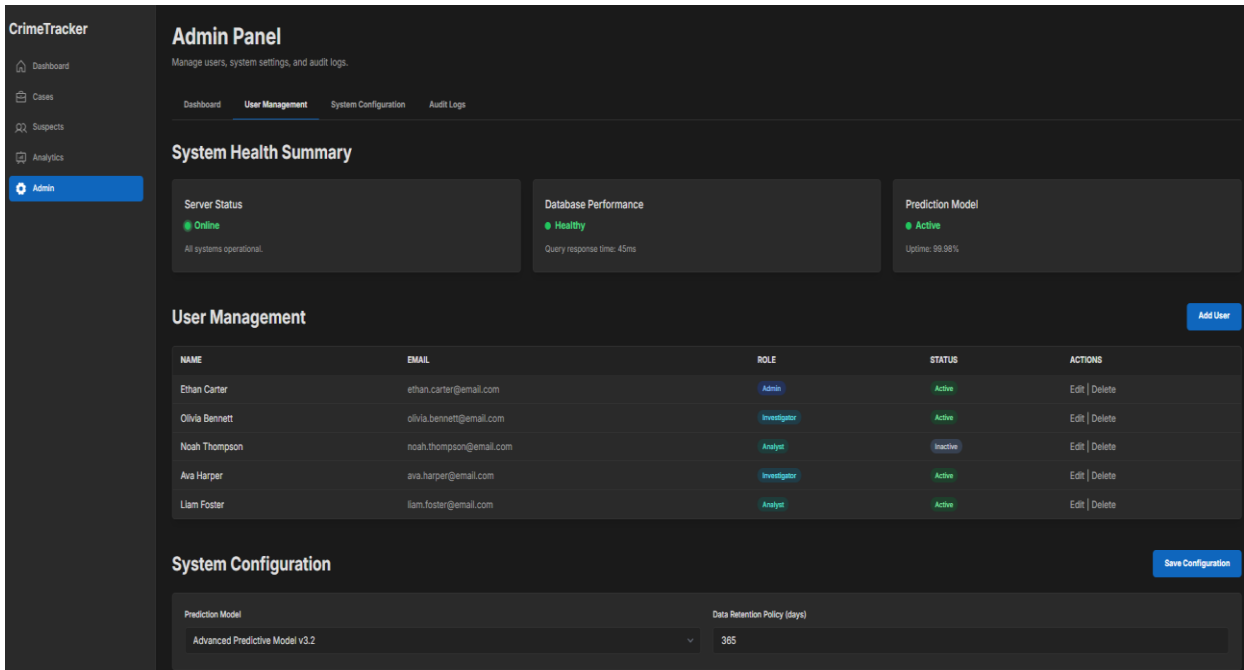


Figure 4.4: Admin Page

- Admin panel is accessible only to authorized users.
- Allows management of cases, officers, and system data.
- Supports adding new entries and assigning officers to cases.
- Enables tracking case progress and generating reports.

- Evidence management

CrimeTracker

Overview

Cases

Evidence

Suspects

Reports

Settings

Evidence Management

Upload, categorize, and manage evidence for cases.

Search evidence by name, type, or case ID

All EvidenceImagesVideosVirtual

FILE NAME	TYPE	DATE	SOURCE	CASE ID	ACTIONS
Security_Footage_01.mp4	Video	2024-03-15	CCTV Camera 4B	CASE-00123	Preview Download
CrimeScene_Photo_A.jpg	Image	2024-03-15	Forensics Unit	CASE-00123	Preview Download
Phone_Records_S1.zip	Virtual	2024-03-16	Digital Forensics Lab	CASE-00456	Preview Download
Witness_Statement_03.pdf	Document	2024-03-17	Detective Miller	CASE-00789	Preview Download
Bodycam_B2.mp4	Video	2024-03-18	Officer Chen	CASE-00789	Preview Download

Audit Trail

TIMESTAMP	USER	ACTION	EVIDENCE ID	DETAILS
2024-03-18 10:05:21	Detective Miller	Access	EVD-001	Viewed file 'Security_Footage_01.mp4'
2024-03-18 09:45:10	Officer Chen	Modification	EVD-005	Uploaded file 'Bodycam_B2.mp4'
2024-03-17 14:20:00	Forensics Unit	Deletion	EVD-002	Deleted file 'Redundant_Photo_X.jpg'
2024-03-16 11:30:45	Digital Forensics Lab	Access	EVD-003	Downloaded file 'Phone_Records_S1.zip'

Figure 4.5: Evidence Management Page

- This page enables investigators to upload, categorize, and manage diverse evidence types—including images, video, and virtual evidence—with metadata, a sortable/searchable gallery, and preview/download options.

- **Behavioral Pattern Analysis**

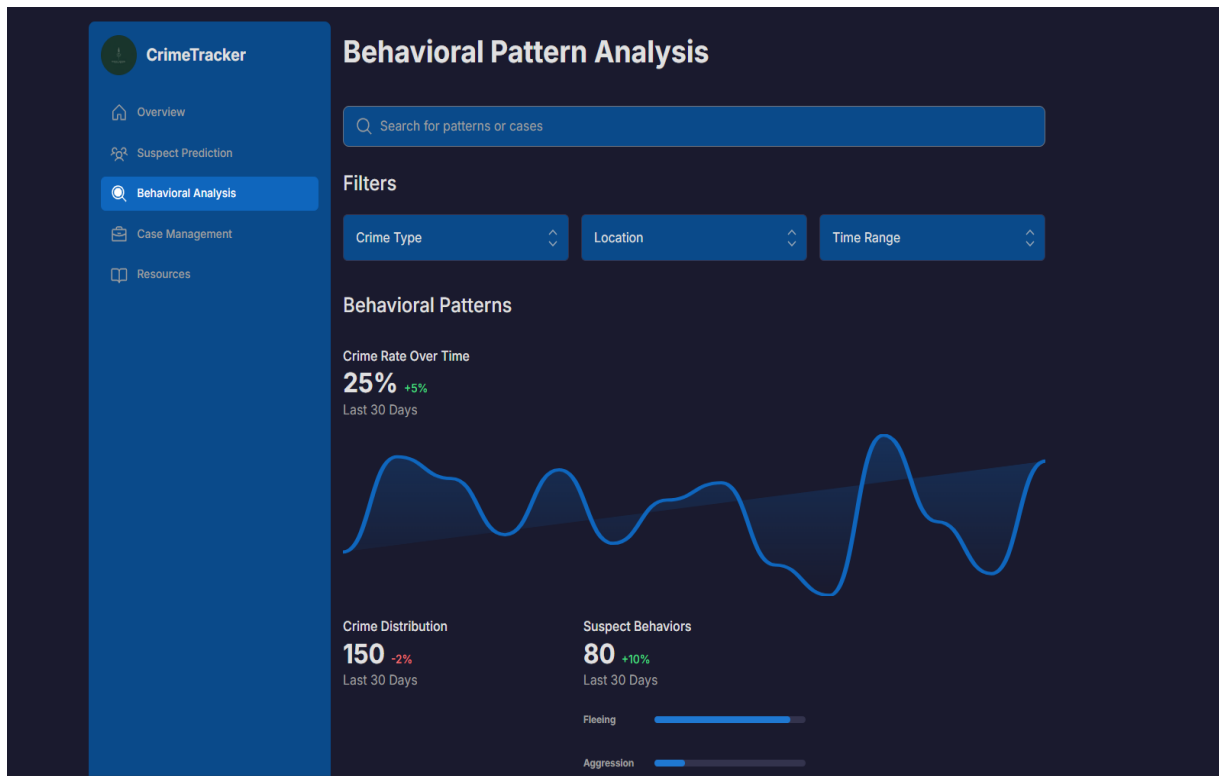


Figure 4.6: Pattern Analysis Page

- Involves analysing data to identify trends in criminal activity.
- Helps predict offender behaviour and proactively identify suspects.
- Aims to anticipate future crimes and support investigative strategies.
- Analysis focuses on factors like type of crime, location, time of day, and past criminal history.

- **Suspect Prediction**

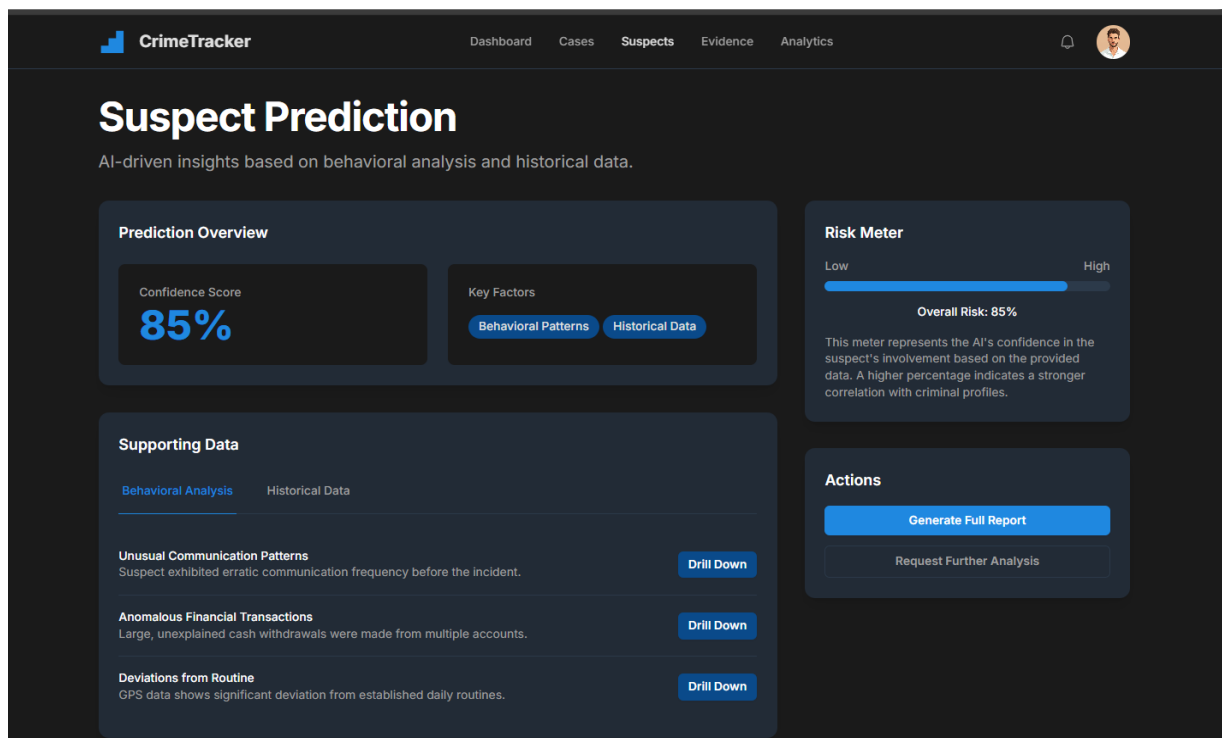


Figure 4.7: Suspect Prediction Page

- This page uses AI to generate suspect predictions with confidence scores and influencing factors, visually presenting results with drill-down capabilities, all within a data-driven

- Alerts and notification

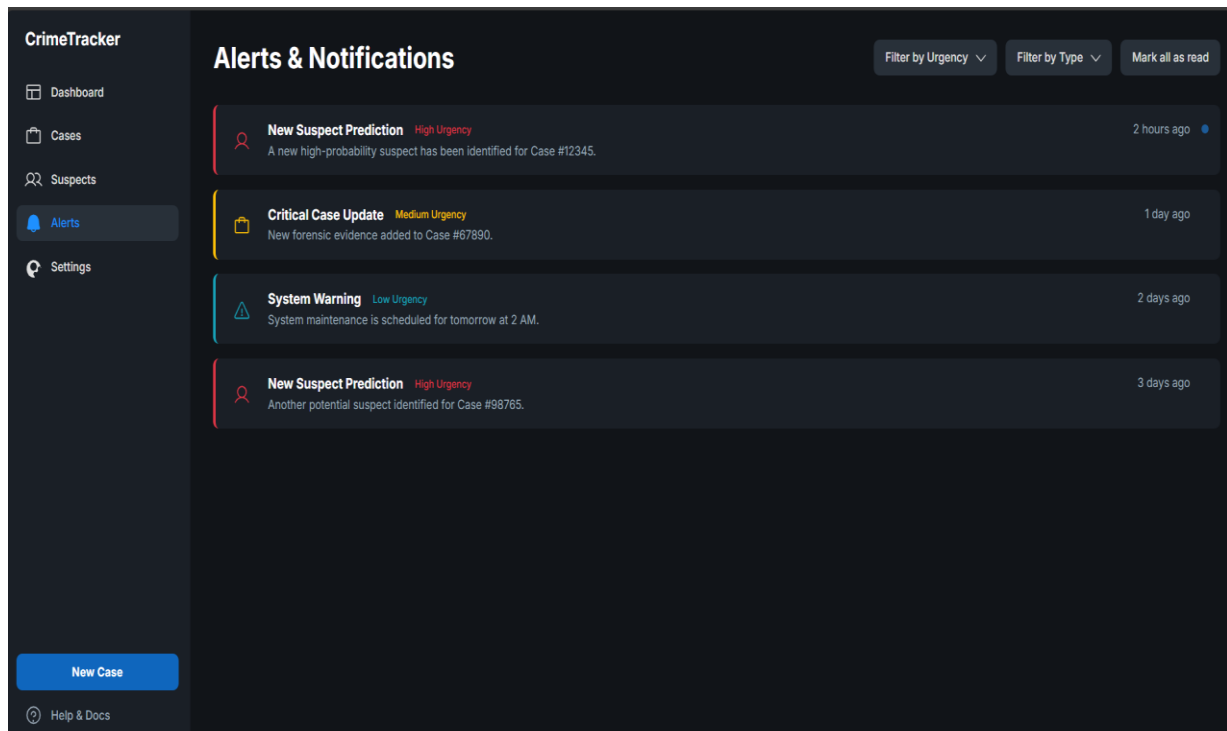


Figure 4.8: Alerts and Notification Page

- Feature designed for police personnel and authorized officials.
- Provides real-time updates and alerts through integrated systems.
- Alerts may include missing persons, wanted criminals, or suspicious activity.
- Enhances situational awareness and response efficiency.

- Investigator access controls

Criminal Tracker

Dashboard

Cases

Investigators

Suspect Prediction

Evidence

Settings

Investigator Access Control

Investigator	Role	Case Access	Suspect Prediction	Evidence Management	Data Visibility
Ethan Carter	Lead Investigator	All Cases	Granted	Granted	Full Access
Olivia Bennett	Junior Investigator	Case 123, Case 456	Revoked	Granted	Limited Access
Noah Thompson	Investigator	Case 789	Granted	Revoked	Partial Access
Ava Rodriguez	Investigator	Case 101, Case 112	Granted	Granted	Full Access
Liam Walker	Junior Investigator	Case 131	Revoked	Revoked	Limited Access

Figure 4.9: Investigator’s Access Page

- Managed through **Role-Based Access Control (RBAC)**.
- Ensures users can only access data relevant to their role and authorization level.
- Key security features include:
 - Secure login modules
 - Multi-factor authentication (MFA)
 - Data encryption

- **Report Generating**

CrimeTracker Dashboard Cases Suspects Evidence Reports Analytics

Report Generation for Case #2024-007

Generate a comprehensive report for the selected case. Customize the content and format to meet your needs.

Report Sections
Select the sections to include in the final report.

- ☒ **Case Details** (ID, Date, Location, Description)
- ☒ **Evidence Log** (Item ID, Description, Location, Status)
- ☐ **Suspect Profiles** (Name, Age, Background, Associates)
- ☐ **Behavioral Analysis** (Risk Assessment, Predicted Actions)
- ☐ **Investigative Timeline** (Key Events, Dates, Outcomes)

Report Format
Choose the export format for your report.

Figure 4.10: Report Generating Page

- This page allows users to create detailed, customizable reports for individual cases, integrating various data points and offering multiple export formats, all within a user-friendly interface and the established dark theme.

4.5 Security Issues

1.Data Privacy and Compliance

- **Issues:** Sensitive data such as Personal identities of suspects and victims, Police record may get leaked and someone may misuse it.
- **Solution:** Proper Encrypting data in database, logs and internal reports. A regularly audit access to manage the data and implementation of pseudonymization technique when training a model.

2. Access Control and Authentication

- **Issues:** Unauthorized users, including internal personnel, may access restricted data or tools.
- **Solution:** Requirement of Multi-factor Authentication for sensitive access points. Monitoring devices and IP-based login anomalies and trigger alerts for unusual activity.

3. AI and Predictive Bias Risks

- **Issues:** AI models may inherit or simplify biases present in historical data, may lead to discriminatory outcomes for the problems.
- **Solution:** Utilizing diverse and balanced datasets to train the predictive models also conducting regular audit checkups for the functionality of algorithmic bias, fairness, and accuracy for the predictions

4. Data Integrity and Tamper Prevention

- **Issues:** Investigative records or AI predictions may be altered intentionally or unintentionally, leading to corrupted case files or false insights.
- **Solution:** Maintain immutable audits and logs of all systems and user activity. Considering hashing data entries and optionally storing them in a blockchain all temple proof ledger.

4.6 Test Cases Design

- Login Page and Authentication Case

Test Case ID	Test Case Name	Required Input	Invalid Output(Rejected)	Valid Output(Accepted)
T1.	Username	Username must not be blank	If wrong/invalid username is given as an input.	If valid/correct username is entered.
T2.	Password	Password must be at least 8 characters long. Also its should contain special characters.	If Password is less than 8 digits.	If Password is correctly matched.
T3.	Authenticating	User enters authenticating Id	If Id is invalid or wrong.	If Id is valid or correct as per the database system.
T4.	Invalid Authenticating	User enters wrong verified Id	Error: “Enter Verified Id for Authentication”	Navigate to Homepage

Table 4.7: Login and Authentication Page Test Case

- **Search /Register New Case**

Test Case ID	Test Case Name	Required Input	Invalid Output(Rejected)	Valid Output(Accepted)
T1.	Search for Suspect	Case Details:- date/time, case history, crime type	Inappropriate or incomplete details	Search results are relevant and appropriate according to given data.
T2.	Add New Case	Details of the Crime Case Filed	If no data is available	Successful addition or updating of new cases
T3.	Add Evidences	Legal evidences and legal report	If data inserted is legal and related to case	Successful evidence insertion.
T4.	Add Other Details	Details related/linked to cases	Inappropriate according to structure	Successful insertion.

Table 4.8: Register New Cases Test Page

- **Evidence Management**

Test Case ID	Test Case Name	Required Input	Invalid Output(Rejected)	Valid Output(Accepted)
T1.	Add evidences	Legal authority permissions.	Inappropriate user or data hampering	Evidences are relevant and recorded successfully.
T2.	Upload Images, CCTV-footages	Details of the Crime scene	If data available should be in stipulated size and format	Successfully Uploaded Evidences.
T3.	View Evidences	Legal authority permissions.	Inappropriate user or data hampering	Successful Evidence View

Table 4.9: Evidence Management Test Page

- **Suspect Prediction**

Test Case ID	Test Case Name	Required Input	Invalid Output(Rejected)	Valid Output(Accepted)
T1.	Add Suspect Details	Valid Suspect related Background information	Inappropriate data related to Suspect	Successful addition of suspect details
T2.	Add Cases History	Details of the Crime and cases	If data available is not relevant.	Successfully Uploaded Evidences.
T3.	Generate prediction	Details related to suspect and crime	Inappropriate data insertion	Successful Prediction View

Table 4.10 : Suspect Prediction Test Page

- **Report Generation**

Test Case ID	Test Case Name	Required Input	Invalid Output(Rejected)	Valid Output(Accepted)
T1.	Add Crime Details	Details of the Crime scene	Inappropriate data for report generation	Report Generated Successfully
T2.	Add Evidence Details	Details of the Evidences available	Inappropriate data for report generation	Report Generated Successfully
T3.	Add Prediction Details	Details of available predictions	Inappropriate data for report generation	Successful Prediction Reports

Table 4. 11: Report Generation Test Page